e-Prime - Advances in Electrical Engineering, Electronics and Energy 4 (2023) 100161

Contents lists available at ScienceDirect

e-Prime - Advances in Electrical Engineering, Electronics and Energy

journal homepage: www.elsevier.com/locate/prime

Transient analysis of power transmission towers above lossy ground with frequency dependent electrical parameters considering the water content of soil☆

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ARTICLE INFO

Keywords:
Transient analysis
Direct lightning
Grounding electrodes
Ground potential rise
Soil modeling

ABSTRACT

Adequate soil modeling is fundamental to assessing the grounding impedance of the grounding system and the transient responses in power systems. This paper aims to investigate three soil models with electrical parameters varying with frequency and water content, namely Smith-Longmire, Scott, and Messier models, and their impact on the ground potential rise (GPR) waveforms of vertical grounding rods. Besides that, the transient induced voltages across the string of insulators induced are also investigated for lightning currents representative first, and subsequent return strokes are employed. The harmonic grounding impedance (HGI) is calculated for rods of 3, 15, and 30 m buried in frequency-dependent (FD) soil model using commercial electromagnetic software (FEKO) with the Method of Moments (MoM) for a range of 100 Hz to 10 MHz. Simulation results show a significant modification of the calculated HGI beyond a particular frequency. The induced transient GPR waveforms show a notable reduction in their voltage peaks when the FD soil model is compared with those computed using the frequency-independent (FI) soil model. Finally, a backflashover analysis is carried out that demonstrates its dependence on the rod length, water content, and the specific soil model employed.

1. Introduction

Tower-footing conductors have an essential role in guaranteeing a safe distribution of the electric potential in transmission towers when they are struck by lightning strikes [1,2]. When lightning hits a power transmission tower, if the tower-footing grounding electrodes are not properly designed, it may lead to insulation failures across the insulator strings by the backflashover (BF) mechanism due to the multiple reflections of surge waves between the top and base of the tower. Then, the BF may lead to outages in the power system during the transient state [3]. As such, the grounding system must present a low impedance to dissipate the surge currents into soil caused by lightning strikes in transmission lines (TLs) [4] and, as a consequence, to decrease the probability of BFs [5].

To correctly compute the impedance of a grounding system, a few factors must be taken into account: (i) arrangement and length of the grounding conductors (vertical, horizontal, grid, etc.); (ii) electrical parameters (resistivity and permittivity) of soil that depend on the frequency and water content; (iii) variation of soil parameters with depth and soil stratification; and (v) ionization phenomenon on the grounding conductors [6–11]. The frequency effect on the soil parameters (resistivity ρ and relative permittivity ε_{rg}) has been extensively investigated in the literature, e.g. [12–15]. The frequency effect notably influences these electrical parameters, especially at high frequencies for highly-resistive soil [12,16]. As a result of this effect, a significant reduction in the transient ground potential rise (GPR) waveform (especially at its peak value) is obtained when a frequency-dependent (FD) soil represents the ground [12,15]. On the other hand, the water content effect on the soil parameters has been considered in [17–21]. (An overview of the models proposed by Scott [17], Smith-Longmire

☆ This study was supported by the São Paulo Research Foundation (FAPESP) (Grants: 2019/01396-1 and 2021/06157-5) and by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - Finance code 001.

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https://doi.org/10.1016/j.prime.2023.100161
Received 2 December 2022; Received in revised form 17 April 2023; Accepted 25 April 2023
Available online 30 April 2023
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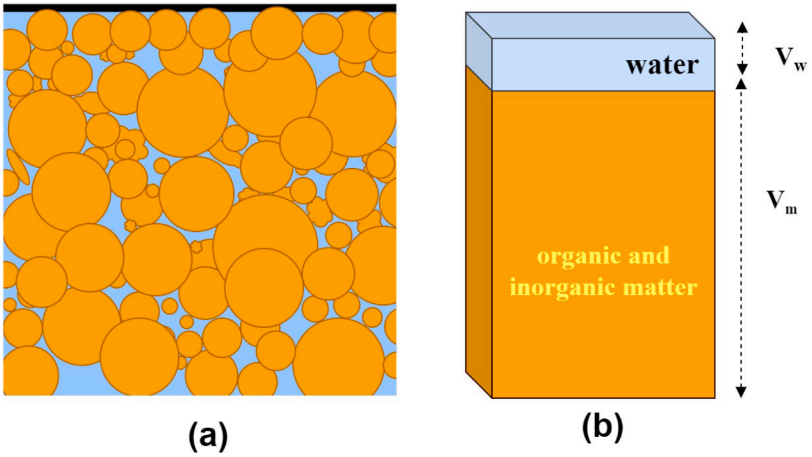


Fig. 1. (a) Saturated soil sample with water; (b) Equivalent percentage volume of each component. (not to scale).

[18], and Messier [19] will be presented in this paper.) In 2003, He et al. [22] studied the influence of the weather conditions (raining or freezing) on the soil concerning the grounding system of an electrical substation, focusing on the induced step and touch voltages. In their analysis, the soil electrical parameters were assumed to be frequency-independent. In [23], Li et al. used a broadband dielectric spectrometer for measuring the soil parameters (resistivity and permittivity) in the frequency range of 1 Hz to 10 MHz using samples of sandy soil. They considered distinct soil samples with a relative humidity level of 0 (dry soil), 1%, and 10% and at −15°C and + 20°C. It is shown that the temperature has little influence on the resistivity when the low water content is present in the soil sample. On the other hand, as the water content increases, its influence is more pronounced. In [11], Datsios et al. conducted an experimental investigation of the lightning impulse behavior of wet sandy soils. In their work, soil samples for a frequency range of 42 Hz up to 1 MHz were analyzed. Further, a time-domain model for the prediction of the impulse impedance with ionization effect was presented, which is suitable for any EMTP-type software. In [24], Ranjan et al. investigated the impact of water content, grain size, and acid rain on the impulse ionization characteristics of sand and bentonite-sand mixture under lightning impulse voltage for a rod electrode buried in these soils. Further, dielectric relaxation spectroscopy was employed to measure the variation of real permittivity and loss tangent for bentonite and bentonite-sand mixture in the frequency range of 0.1 Hz to 10 MHz at temperatures of 30, 50 and 70°C.

Recently, Nazari et al. [21] used a model to solve the Electric Field Integral Equation (EFIE) on the grounding system, composed of long rods, of a wind turbine. For this analysis, the EFIE is solved using the Method of Moments (MoM). For the electromagnetic analysis, the model of Messier [19] was adopted. Salarieh et al. [25], carried out an analysis and comparison of four models (Smith-Longmire [26], Scott [17], Messier [19], and Datsios [27]) considering the soil moisture content and lightning performance on cylindrical electrodes. In their analysis, only vertical and horizontal electrodes were studied whose grounding impedance was calculated using the Finite Element Method (FEM) in the frequency range of 1 kHz to 5 MHz. The induced transient GPR in each grounding electrode was also investigated. The results demonstrated significant differences on the GPR waveforms for the soil models adopted.

Concerning the impedance of grounding systems, several methods are available in the literature that can be organized as follows: (i) Circuit theory based methods using either lumped-element [28,29] or distributed-parameter models [30]; and (ii) Numerical electromagnetic models (EM) that solve full Maxwell's equations. In this context, numerical methods based on the Finite Difference Time Domain (FDTD) Method [31], Method of Moments (MoM) [32,33], Finite Element Method (FEM) [34], or Partial Element Equivalent Circuit (PEEC) [35] have been employed. The EM models are assumably more accurate, however they are computationally costly, depending on the grounding system arrangement and mesh size of the simulation arrangement [34]. Nevertheless, the previously-mentioned works do not consider how the water content affects the induced overvoltage across the string of insulators and its impact on the lightning performance focusing on the soil models available in the literature.

This paper investigates the models proposed by Scott [17], Smith-Longmire [18], and Messier [19] and their impact on the lightning-induced transient GPR for grounding rods and on the voltages across the string of insulators of a transmission tower. The harmonic grounding impedance (HGI) for three vertical rods of 3, 15, and 30 m buried in soils with frequency-dependent and water-variable electrical parameters is calculated using a full-wave electromagnetic software, FEKO [36], in the frequency range of 100 Hz to 10 MHz. The transient responses are calculated for lightning currents representative of the first and subsequent return strokes. All transient responses are compared with those obtained, assuming a frequency-independent (FI) soil model. Simulation results demonstrated that a significant difference is observed for HGI above a particular frequency, and especially at high frequencies, between the soil models. As a result, a noticeable reduction in the transient GPR waveforms for the rods is seen in comparison with those waveforms obtained using a frequency-independent soil model. Concerning the voltages across the upper string of insulators of a tower structure with grounding rods buried in a highly-resistive soil, the results have indicated that different responses (peak values and waveforms) are obtained using the three soil models in comparison to those assessed using the frequency-independent (FI) soils. Furthermore, the backflashover occurs at different times depending on the soil model used for the study.

This paper provides a comparison between three frequency-dependent (FD) models and frequency-independent (FI) models of soil available in the literature. This study investigates the impact of different soil models on the harmonic impedance, ground potential rise (GPR), and induced overvoltages across a string of insulators which determines the probability of a backflashover. It is demonstrated that the harmonic grounding impedance is significantly affected by the soil model. Consequently, the generated GPR, overvoltages across the string of insulators, and the occurrence of the backflashover not only depend on the injected lightning current characteristics but also on the soil model adopted. To the best of our knowledge, such extensive study has not been presented in the literature before, concerning grounding rods at the tower footing of transmission lines, focusing on the induced voltages on the string of insulators. The novelty of this study lies on the lightning performance of transmission lines (tower structure with grounding rods)

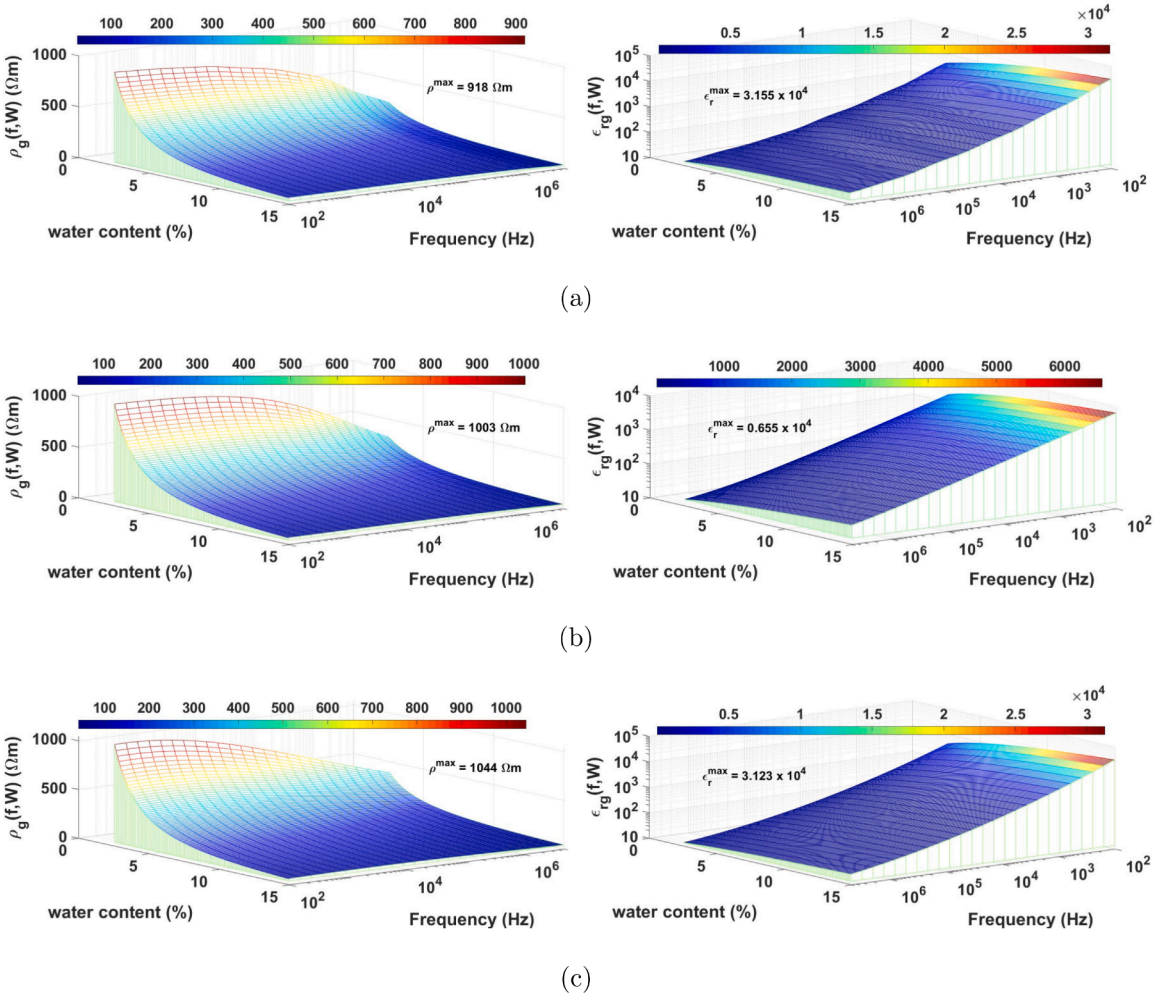


Fig. 2. Resistivity (left-hand side column) and relative permittivity (right-hand side column) computed using models of: (a) Smith-Longmire, (b) Messier, and (c) Scott.

on more realistic grounds including the water content as a physical factor, proposed by Scott [17], Smith-Longmire [18], and Messier [19].

The main contributions of this paper include the analysis of the impact of the water content on the harmonic grounding impedance of rods and the overvoltages generated on transmission lines subjected to lightning strikes. Further, the backflashover analysis demonstrated that outages might occur in power systems depending on the soil modeling, showing the importance of considering the water content in the transient studies.

The rest of the paper is organized as follows: Section 2 describes closed-form expressions for three soil models depending on the frequency and water content. Section 3 presents the simulation model and the methodology to compute the transient responses. In Sections 4 and 5, the numerical results concerning the backflashovers are depicted and discussed. Finally, in Section 6, the main conclusions of the paper are summarized.

2. Modeling of electrical parameters of soil

Soil is composed of four components, including a solid fraction formed by inorganic matter (minerals) and organic matter (from plants and animals), a liquid fraction formed by water (aqueous solutions), and gases filling the voids. For the saturated condition (all voids are filled with water), a generic representation of the soil sample is illustrated in Fig. 1 [37]. The volumetric water content of a bulk of soil is represented as the following ratio

$$W(\%) = \frac{V_w}{V_w + V_m} \times 100\% \tag{1}$$

where V_w and V_m are the volumes of water and solid matter (organic and inorganic), respectively. In terms of electromagnetic properties, the soil is modeled by the magnetic permeability (μ_g), electric resistivity (ρ_g), and relative permittivity (ε_{rg}), where the last two are strongly variant with frequency, and water content [15]. The frequency effect on the soil electrical parameters occurs due to the several polarization processes in the ground molecules as the frequency increases [15].

The frequency effect has practically no influence on the soil magnetic permeability (μ_g) where, for practical engineering applications, the magnetic permeability of free space (μ_0) is considered. The soil becomes a semiconductor as the water content increases due to the dissolved ions that result in a high ε_{rg}, much higher than that of dry soil. On the other hand, the soil resistivity (ρ_g) decreases as the soil becomes wetter. Based on measurements conducted in the field and laboratory with distinct soil samples, different expressions have been proposed in the literature, such as [17–19,27]. In this section, three closed-form expressions proposed by Scott [17], Smith-Longmire [18], and Messier [19] are briefly presented.

(i) Scott (S) Model

Scott et al. [17] investigated 105 rock and soil samples with a variation of 0.3% to 82% of moisture content [17]. The formulas

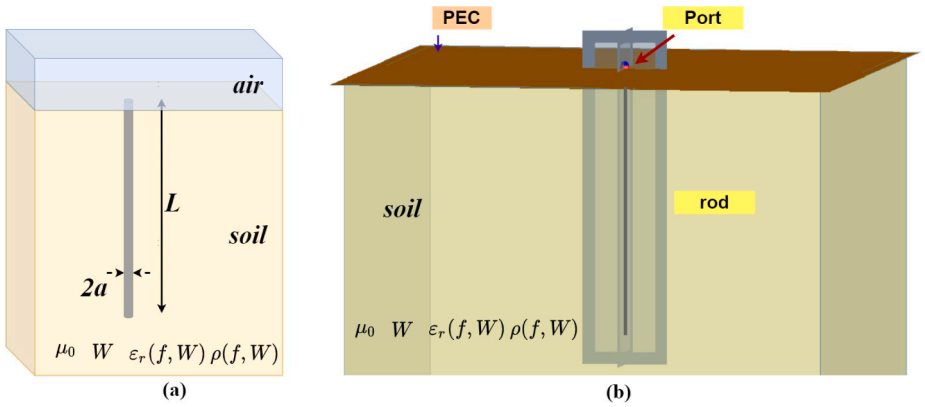


Fig. 3. (a) Rod buried in the FD soil; (b) Simulation arrangement in FEKO.

to estimate the resistivity and relative permittivity of the soil were obtained based on the measured impedance of the soil samples in the laboratory [15]. These fitted closed-form expressions to estimate the resistivity (in kΩm) and relative permittivity are given by Scott et al. [17]

$$\rho_g(f,W) = 10^{0.604-1.64W_1+0.062f_1-0.062W_1^2+0.07W_1f_1-0.021f_1^2} \tag{2}$$

$$\varepsilon_{rg}(f,W) = 10^{4.905+1.308W_1-0.971f_1+0.111W_1^2-0.168W_1f_1+0.059f_1^2}. \tag{3}$$

In (2) and (3), $f_1 = \log_{10}(f)$ and $W_1 = \log_{10}(W)$ where f is the frequency in Hz and W is the percentage of moisture content.

(ii) Smith-Longmire (SL) Model

Based on the data from Scott et al. [17], Smith and Longmire [18] developed the first semi-theoretical model to represent the resistivity and relative permittivity for a range of 100 Hz to 200 MHz which was named the *Universal Soil Model* being valid for the volume percent of water between 0.2 to 35% [15,18]. The closed-form formulas for the resistivity and relative permittivity are calculated using [15]

$$\rho_g(f,W) = \left[\beta_0 + \varepsilon_0 \sum_{n=1}^{13} a_n \beta_n \frac{(\omega/\beta_n)^2}{1+(\omega/\beta_n)^2}\right]^{-1} \tag{4}$$

$$\varepsilon_{rg}(f,W) = \varepsilon_\infty + \sum_{n=1}^{13} \frac{a_n}{1+(\omega/\beta_n)^2} \tag{5}$$

where $\omega = 2\pi f$ (rad/s) is the angular frequency, ε_∞ is the permittivity at the high frequency limit ($\varepsilon_\infty = 5$) and ε_0 is the vacuum permittivity ($\varepsilon_0 = 8.854\times10^{-12}$ F/m). The coefficients $\beta_0 = 8\times10^{-3}(W/10)^{1.54}$ and $\beta_n = (W/10)^{1.28}10^{n-1}$, and the coefficients a_n are specified in the Table 3.1 of [15]. Furthermore, the proposed model establishes the causal relationship as ruled by Kramers-Kronig's relation as detailed in [15].

(iii) Messier (M) Model As an alternative model for the equations from Scott's data, Messier [19] generated an empirical model that represents the resistivity and relative permittivity for a range of 100 Hz to 10 MHz. These expressions are given by Messier [19]

$$\rho_g(f,W) = \frac{\rho_{DC}}{1+\sqrt{4\pi f\varepsilon_\infty\rho_{DC}}} \tag{6}$$

$$\varepsilon_{rg}(f,W) = \frac{\varepsilon_\infty}{\varepsilon_0}\left(1+\sqrt{\frac{1}{\pi f\varepsilon_\infty\rho_{DC}}}\right) \tag{7}$$

where $\varepsilon_\infty = 8\varepsilon_0$, is the relative permittivity at higher frequencies and $\rho_{DC} = [8\times10^{-3}(W/10)^{1.54}]^{-1}$. As described in [15], these closed-form expressions also obey the Kramers-Kronig'relations, resulting in a causal model.

2.1. Comparison of the soil models

The resistivity $\rho_g(f, W)$ and the relative permittivity $\varepsilon_{rg}(f, W)$ proposed by each of Scott (S), Smith-Longmire (SL), and Messier (M) models are plotted as surface graphs in Fig. 2 in the range of 100 Hz to 10 MHz and for the water content of 2.5% to 15%. As seen in this figure, in general, soil resistivity value is notably dependent on water content and frequency, reducing as the water content increases, which is more pronounced at high frequencies. The low-frequency resistivity in the Smith-Longmire (SL) is the lowest compared to those calculated by the other two approaches. Regarding the permittivity $\varepsilon_{rg}(f, W)$, one can see that the permittivity increases for the increased moisture content and decreases for the increasing frequency associated with the different polarization mechanisms in the soil molecules [15]. It is worth mentioning that the Smith–Longmire's and Messier's approaches are causal, whereas the Scott model does not satisfy this condition [14]. However, Scott's model led to reasonable simulation results in the transient GPR waveforms developed for cylindrical electrodes (vertical and horizontal) [20] as well as those presented in this work. The dependence on the water content and frequency will impact the HGI of the grounding rods and the time-domain responses, as presented in the following sections.

3. Methodology

The harmonic grounding impedance (HGI) of a vertical rod buried in the lossy ground is calculated using a commercial electromagnetic software FEKO (*in full-wave approach*) employing the Method of Moments (MoM) [36]. The metallic object (in this case, the buried rod) is modeled as an antenna immersed in a dielectric medium of finite conductivity (soil), considering the frequency- and moisture-dependent of soil electrical parameters. To calculate the HGI of the metallic rod, an infinite perfect electric conductor (PEC) plane at the boundary of air and soil is used, as shown in Fig. 3. A numerical port is established between the PEC surface and the top of the rod and a voltage difference of $V_s(j\omega) = 1\angle0°$ V is applied between its two terminals, inducing the current $I_{in}(j\omega)$ to flow into the rod.

In this figure, a Perfect Electric Conductor (PEC) plane is on the top of the homogeneous soil where the grounding electrode is buried. The ground is represented by its frequency-dependent (FD) and water variable electrical parameters [μ_0, $\varepsilon_r(f, W)$, and $\rho_g(f, W)$] and also by its frequency-independent (FI) electrical parameters [μ_0, ε_r, and ρ_g]. The validity of the simulation arrangement used to compute the harmonic grounding impedance here has already been verified in [38,39]. The frequency-dependent HGI can be computed as

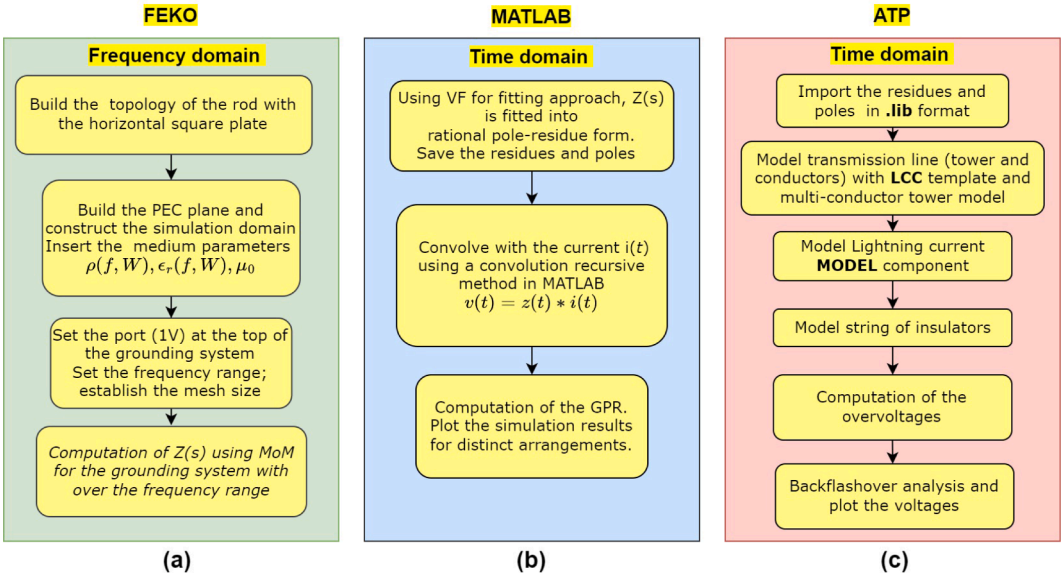


Fig. 4. Flow-cart with the steps to compute: (a) harmonic grounding impedance of vertical electrodes; (b) transient GPR; (c) overvoltages and backflashover analysis.

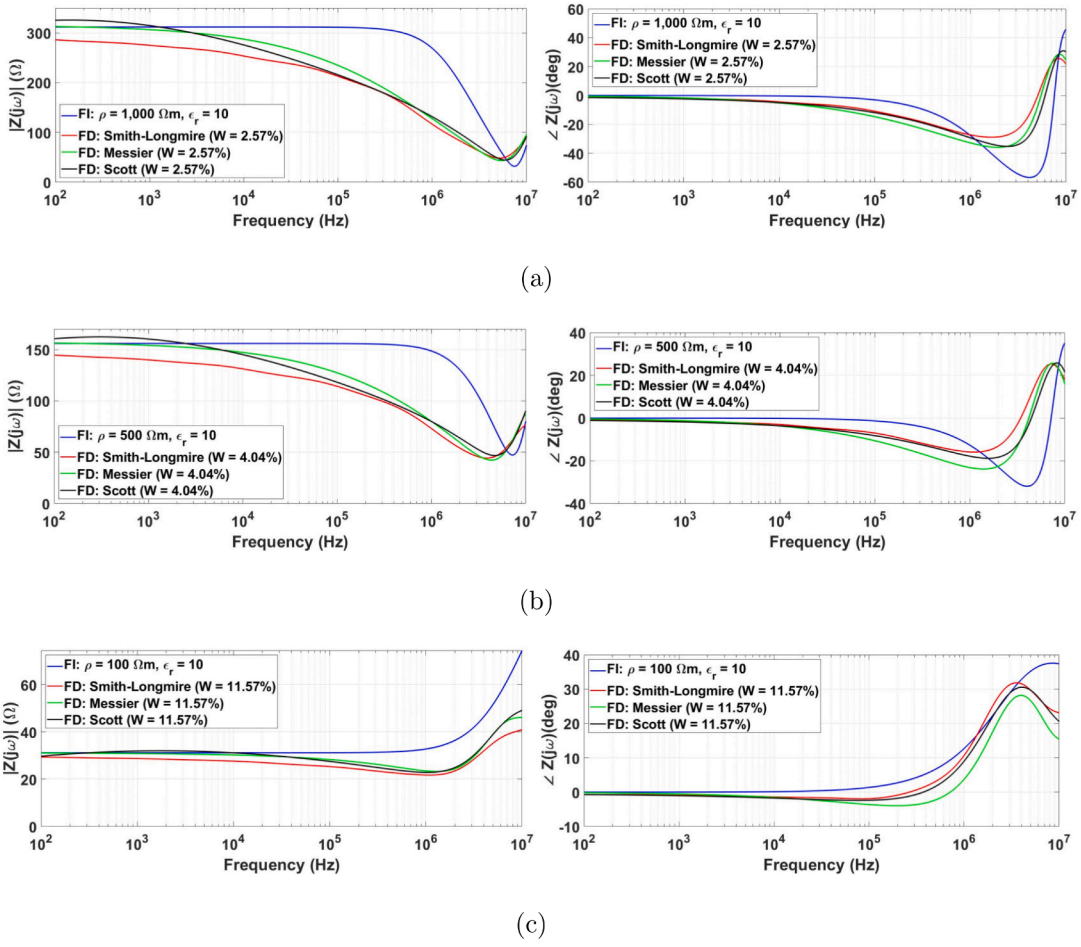


Fig. 5. Magnitude (left-hand side column) and phase (right-hand side column) of the HGI for 3-m rod assuming a water content (W) of (a) 2.57%, (b) 4.04%, and (c) 11.57%.

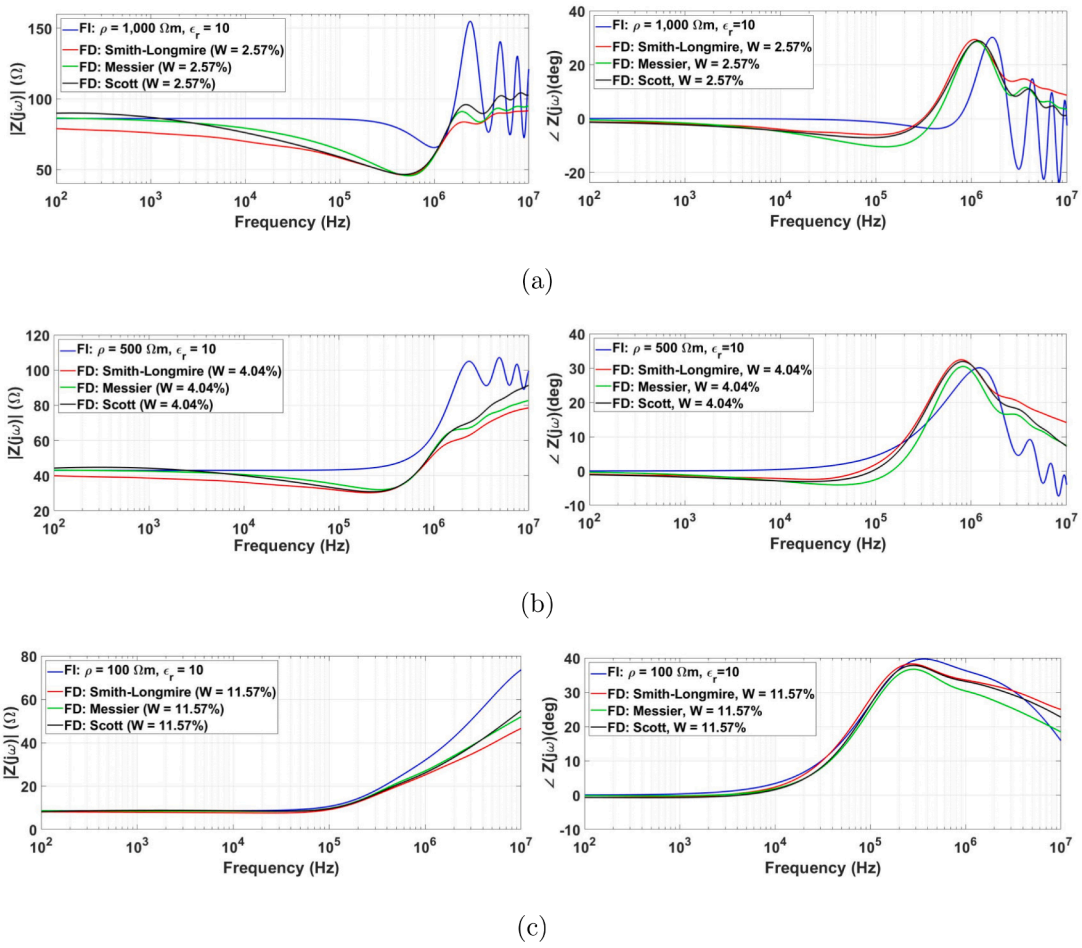


Fig. 6. Magnitude (left-hand side column) and phase (right-hand side column) of the HGI for 15-m rod computed with water content (W) of (a) 2.57%, (b) 4.04%, and (c) 11.57%.

$$Z_h(j\omega) = V_s(j\omega)/I_{in}(j\omega). \tag{8}$$

The HGI $Z_h(j\omega)$ is approximated by a function $Z_{fit}(s)$, which consists of a sum of rational functions given generically by Gustavsen and Semlyen [40]

$$Z_h(s) \approx Z_{fit}(s) = \sum_{j=1}^{N}\left(\frac{r_j}{s+p_j}\right) + d + bs, \tag{9}$$

where r_j are the residues, p_j are the poles (real or complex conjugate pairs), the terms d and b are constants for each fitting procedure, N is the number of poles for the rational synthesis, and $s = j\omega$ is the complex frequency in rad/s [39,40]. The values of r_j, p_j, d and b are calculated using the Vector Fitting (VF) method assuming a certain number of poles N [1,40,41]. The transient GPR $v(t)$ generated for a current $i(t)$ injected at the feed point, from a remote point, is expressed as follows

$$v(t) = \mathcal{L}^{-1}\{Z_{fit}(s) \times \mathcal{L}\{i(t)\}\} = z_{fit}(t) * i(t), \tag{10}$$

where $\mathcal{L}$ is the Laplace transform and (*) represents the convolution between the two functions. The GPR $v(t)$ is computed using a recursive convolution method given by

$$v(t) = \alpha v(t-\Delta t) + \beta i(t) + \mu i(t-\Delta t), \tag{11}$$

where coefficients α, β and μ are obtained assuming that the injected current $i(t)$ is linear in the small time-step Δt. The coefficients are detailed in [42]. A flow-cart summarizing each step to compute the harmonic grounding impedance illustrated in Fig. 4a.

4. Numerical results

4.1. Harmonic grounding impedance of the grounding rods

The HGI of the three vertical rods buried in soil with parameters depending on the frequency and water content represented by Smith-Longmire (SL), Scott (S), and Messier (M) models is calculated using the full-wave electromagnetic software FEKO (Fig. 3b). In these simulations, the HGI is computed in the frequency range of 100 Hz to 10 MHz for three values of water content (W) of 2.57%, 4.04%, and 11.57%. These values of the water content corresponding to a low-frequency soil resistivity of approximately 1000, 500, and 100 Ωm, respectively, obtained by (2), (4), and (6). The values of 3, 15, and 30 m are considered for the length of the rods (L), and they all have a radius of $a = 12.5$ mm. Additionally, the HGI of the rod is compared with that obtained assuming a frequency-independent soil (*i.e.* constant ρ_g and ε_{rg}). The relative permittivity of $\varepsilon_{rg} = 10$ was assumed in all frequency-independent soils.

The HGI of the rods is illustrated in Figs. 5–7. According to these figures, the HGI is considerably modified, especially above a particular frequency, when the frequency effect and the water content are taken into account by Scott, Smith-Longmire, and Messier models, compared to those assessed assuming a frequency-independent (FI) soil model (blue lines). These modifications are more pronounced when the rods are buried in highly-resistive soils (low water content of 2.57%), as shown in Figs. 5a, 6a, and 7a. However, the proposed models of Scott, Smith-Longmire, and Messier tend to present similar values of the HGI when a high water content (W = 11.57%) is considered, as shown in

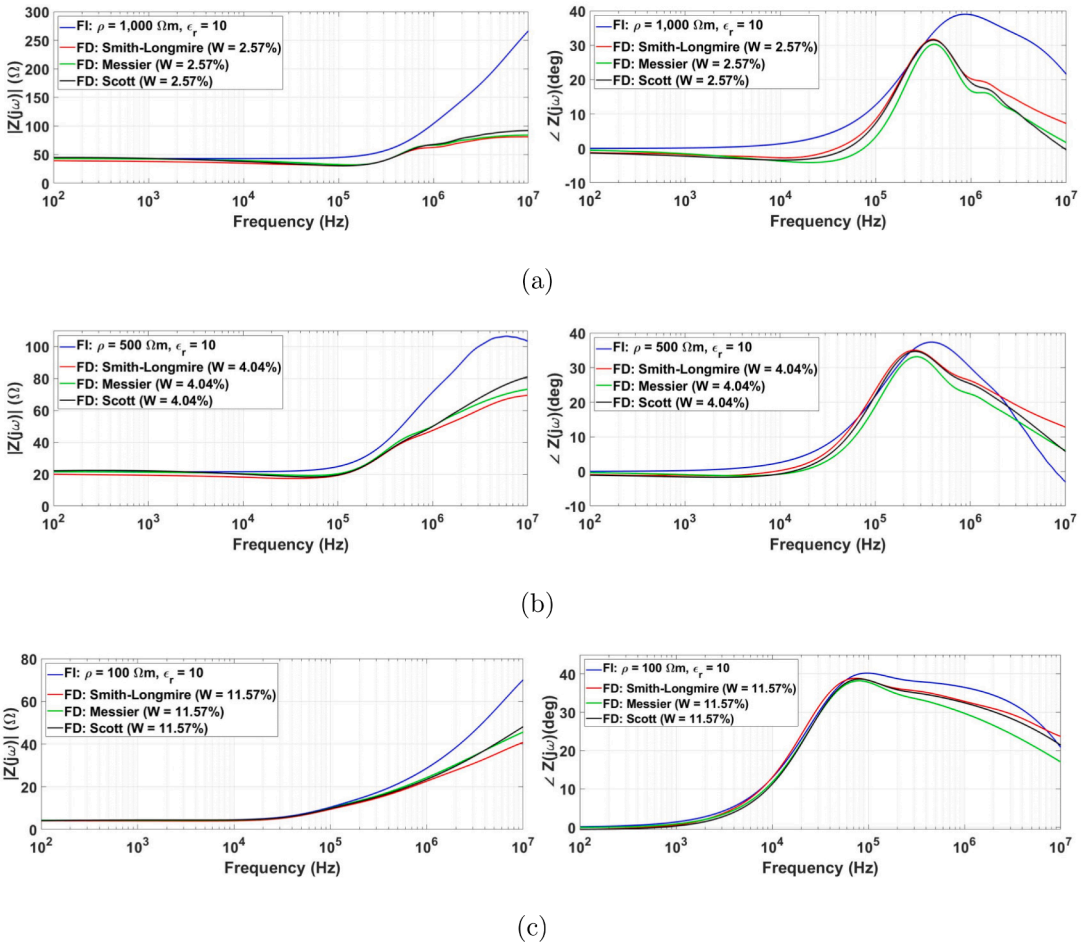


Fig. 7. Magnitude (left-hand side column) and phase (right-hand side column) of the HGI for 30-m rod computed with water content (W) of (a) 2.57%, (b) 4.04%, and (c) 11.57%.

Table 1
Comparison of low-frequency resistance $R_{LF}(\Omega)$ calculated at 100 Hz using the frequency-independent (FI) and three frequency-dependent (FD) models of Smith-Longmire (SL), Messier (M), and Scott (S) for different values of water content W and lengths of the rod L.

	W= 2.57%				W= 4.04%				W=11.57%			
L(m)	FI soil	SL	M	S	FI soil	SL	M	S	FI soil	SL	M	S
3	312	286	313	325	156	145	156	160	31	29	31	30
15	86	79	86	90	43	40	43	44	9	8	9	8
30	43	39	43	45	22	20	22	22	4	4	4	4

Figs. 5c, 6c, and 7c. Above a particular frequency, the so-called characteristic frequency, the HGI becomes more reactive where either an inductive or a capacitive predominance in the impedance is seen depending on the frequency range. The geometrical parameters (rod length, radius) and the electrical parameters (soil resistivity, moisture content) significantly affect the harmonic impedance and the

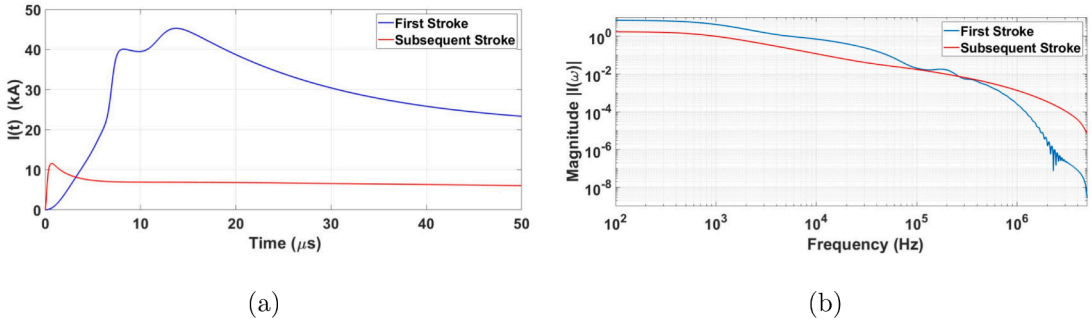


Fig. 8. Lightning current waveforms calculated using (12) for parameters adopted from De Conti and Visacro [43]: (a) time-domain and (b) frequency-domain (magnitude of the Fourier Transform).

Table 2
Comparison of the lightning parameters for first and subsequent return strokes.

Lightning Current	I_p (kA)	T10 (μs)	T30 (μs)	T50 (μs)	S10 (kA/μs)	S30 (kA/μs)	S_{max}(kA/μs)
First	45.3	5.2	3.0	53.8	6.2	8.0	20.2
Subsequent	12.0	0.5	0.30	48.4	26.8	24.5	40.1

characteristic frequency of the electrode [20]. Moreover, for both frequency-dependent or frequency-independent (FI) models of soil, the HGI presents a resistive behavior essentially at low frequencies. This resistance is the low-frequency resistance (R_{LF}), which is proportional to the ground resistivity and decreases with increasing the electrode length. The values of R_{LF} for the three frequency-dependent and the frequency-independent soil models are summarized in Table 1. This table shows as the water content and electrode length increase, R_{LF} values calculated by different models are similar. However, highly-resistive soil (low water content) shows differences between the frequency-dependent soil models compared to those from the frequency-independent model of soil. These differences are due to significant variation of the low-frequency resistivity (ρ_0) calculated at 100 Hz, which is denoted as the maximum resistivity (ρ^{max}) in Fig. 2a.

4.2. Transient ground potential rise dependence on soil model

To evaluate the influence of the FD soil electrical parameters with variable water content on the performance of transient voltages, the grounding rods' transient ground potential rise (GPR) will be analyzed in this section. The transient GPR generated for each lightning current (first and subsequent return strokes) is calculated using the procedure outlined in Section 3. Each lightning strike is modeled as an impulsive current source applied at the rod's tip. The current waveform is given by De Conti and Visacro [43]

$$I(t) = \sum_{k=1}^{m} \frac{I_{0k}}{\eta_k} \frac{(t/\tau_{1k})^{n_k}}{1 + (t/\tau_{1k})^{n_k}} e^{-t/\tau_{2k}} \tag{12}$$

where

$$\eta_k = \exp\left[-\left(\frac{\tau_{1k}}{\tau_{2k}}\right)\left(n_k \frac{\tau_{2k}}{\tau_{1k}}\right)\right]^{1/n_k}.$$

where I_{0k} is the current peak, τ_{1k} is the front time, τ_{2k} is the decay time, n_k is an integer coefficient, and η_k is an exponent related to the steepness of the kth current waveform. These parameters are detailed in [43]. The time-domain current waveforms and the magnitude of its Fourier Transform are shown in Fig. 8. Based on the Fig. 8(a), in order to compare the lightning first and subsequent return strokes, the Table 2 depicts five parameters, detailed as follows:

- *Ip*: peak value of the current, in kA;
- *T*10, *T*30: time for the lightning current to increase from 10% to 90% and 30% to 90% of the first peak, respectively, in μs;
- *T*50: time related to the current decay 50% of the peak value, in μs;

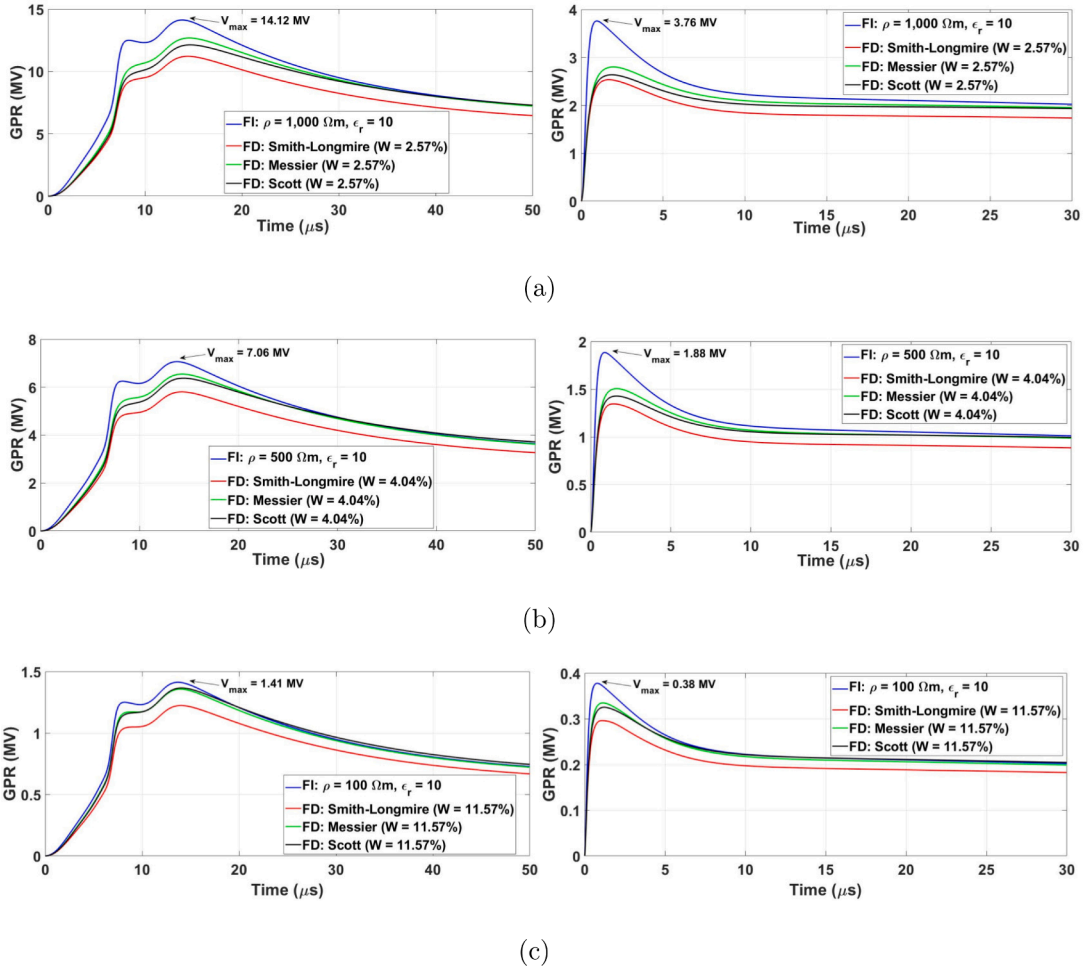


Fig. 9. Transient GPR waveforms for the first lightning return stroke (left-hand side column) and subsequent return stroke (right-hand side column) for the 3-m rod computed with water content (W) of (a) 2.57%, (b) 4.04%, and (c) 11.57%.

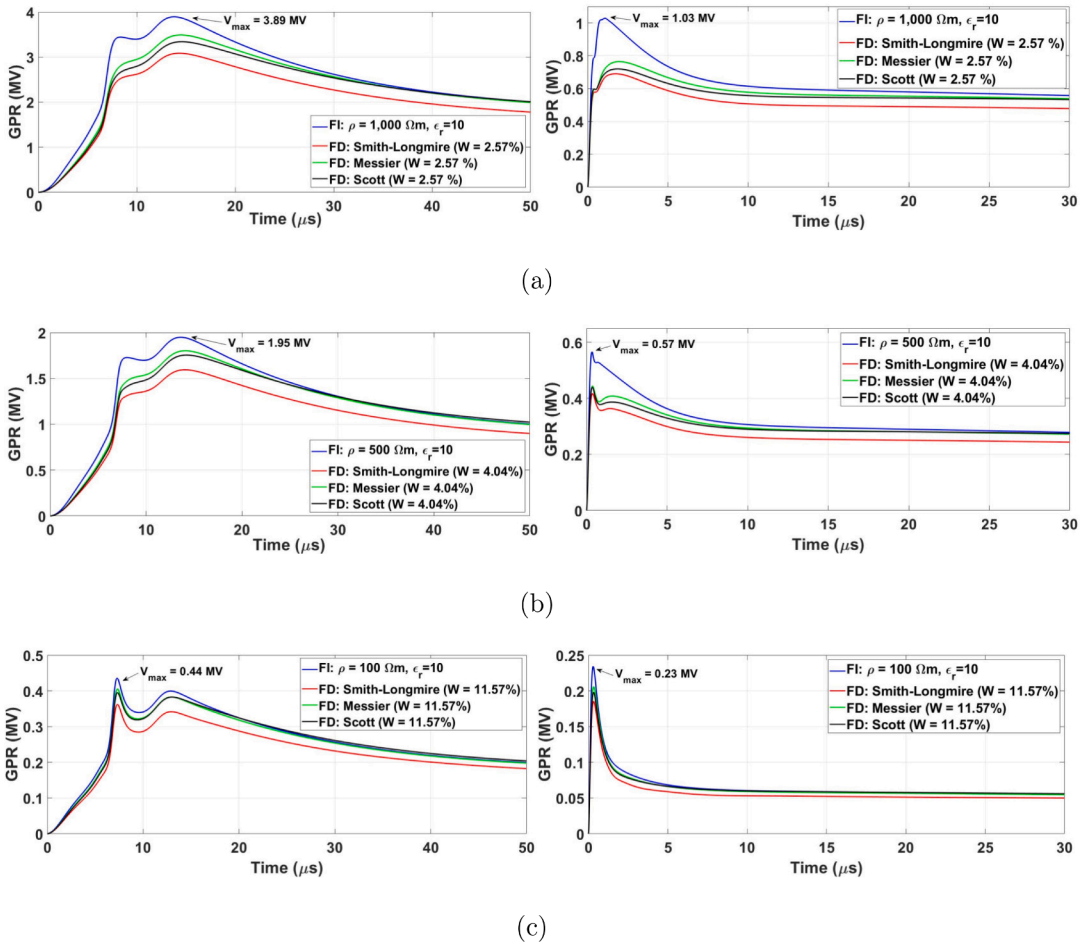


Fig. 10. Transient GPR waveforms for the first lightning return stroke (left-hand side column) and subsequent return stroke (right-hand side column) for 15-m computed with water content (W) of (a) 2.57%, (b) 4.04%, and (c) 11.57%.

- $S10$, $S30$: steepness of a straight line from 10% to 90% and 30% to 90% of the current, respectively, in kA/μs;
- S_{max}: maximum time derivative of the lightning current (di/dt), kA/μs.

According to this table, the current peak of subsequent is around three times lower than that from first return stroke, and the times T10 and T30 are much smaller. However, the $S10$, $S30$ and S_{max} are much higher for the subsequent return stroke. These parameters affects the Fourier transform plotted in Fig. 8(b), where at low frequencies, the first stroke has higher amplitude, however, at higher frequencies, the magnitude of the subsequent stroke is greater. This behavior impacts significantly the waveforms of the GPR and voltages at the tower, as will be seen in the following sections.

A flow-cart summarizing each step to compute the GPR is illustrated in Fig. 4b. The number of poles to fit the harmonic impedance varies between 20 to 30 for the HGI. A time step of 10 ns is assumed to compute the time-domain responses using the convolution recursive method [42]. The transient GPR waveforms computed for the 3, 15, and 30-m long rods buried in the frequency-independent soil and the three frequency-dependent soil models are plotted in Figs. 9, 10, and 11, respectively. It can be seen that the transient GPR waveforms obtained with the frequency-independent soil model present the highest voltage peaks ($V_{\max}$) for all time-domain simulations for both types of lightning return stroke currents. Furthermore, a significant reduction is observed in the peak values of the transient GPR waveforms when a frequency-dependent soil with variable water content is assumed. These reductions are more pronounced when the subsequent return stroke is injected in the grounding rods buried in highly-resistive soil (*i.e.* low water content). These pronounced reductions occur due to the higher frequency spectrum in the subsequent lightning return stroke current (shorter front time). The frequency effect on the soil electrical parameters (resistivity and permittivity) is more significant for the case of the subsequent lightning return stroke, resulting in a higher difference between the transient GPR calculated using the frequency-independent (FI) soil model and that calculated using the frequency-dependent soil model [21]. Considering a certain electrode length, as the water content increases from 2.57% to 11.57%, the voltage peak is reduced by up to 10 times, which reveals the influence of the water content on the modeling of soil.

To study the impact of the FD soil models on the performance of GPR, the relative reduction $\delta(\%)$ of peak transient GPR using the FD soil models are calculated as given by

$$\delta(\%) = \frac{V_p^{FI} - V_p^{FD}}{V_p^{FI}} \times 100\% \tag{13}$$

where V_p^{FI} and V_p^{FD} represent the peak of the transient GPR waveforms for the frequency-independent and frequency-dependent soil models, respectively. Table 3 shows the voltage peak (in MV) and percentage variation $\delta(\%)$ for each case. As seen in this Table, soil with less water content (W=2.57 %) will present higher peaks and percentage reduction for a certain electrode length. Additionally, this table depicts that the influence of frequency effect on the soil parameters is more significant for the subsequent return stroke due to its higher frequency content [21].

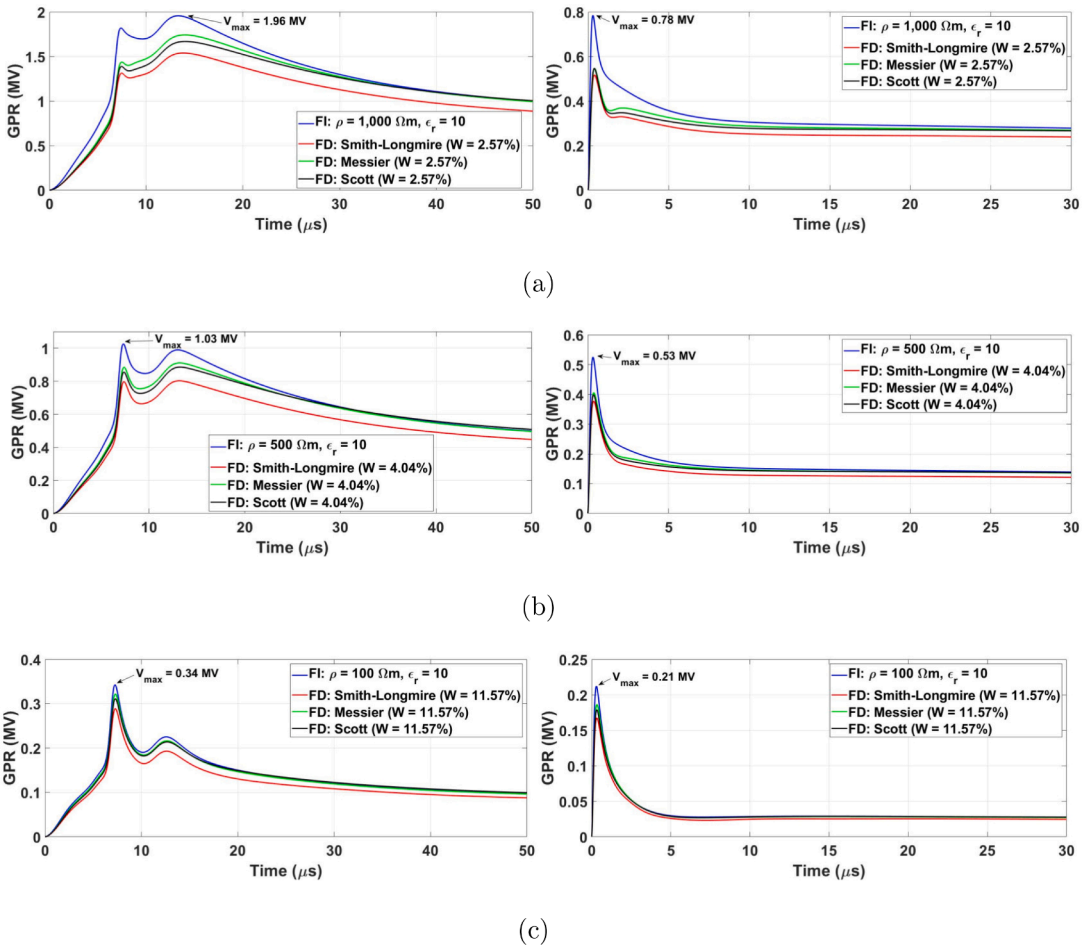


Fig. 11. Transient GPR waveforms for the first lightning return stroke (left-hand side column) and subsequent return stroke (right-hand side column) for the 30-m rod computed with water content (W) of (a) 2.57%, (b) 4.04%, and (c) 11.57%.

Table 3
Peak of the transient GPR (in MV) calculated for the frequency-independent (FI) and three models of Smith-Longmire (SL), Messier (M), and Scott (S) for different values of water content W(%) and lengths of the rod L. The number in () represents the percentage variation calculated using (13).

	Lightning First Return Stroke					Lightning Subsequent Return Stroke				
L	FI soil	W(%)	SL	M	S	FI soil	W(%)	SL	M	S
3m	14.12(0)	2.57	11.21(21)	12.68(10)	12.13(14)	3.76(0)	2.57	2.53(33)	2.80(26)	2.64(30)
	7.06(0)	4.04	5.80(18)	6.54(7)	6.37(10)	1.88(0)	4.04	1.34(29)	1.51(20)	1.43(24)
	1.41(0)	11.57	1.22(13)	1.36(4)	1.37(3)	0.38(0)	11.57	0.30(22)	0.34(11)	0.33(14)
15m	3.89(0)	2.57	3.09(21)	3.49(10)	3.34(14)	1.03(0)	2.57	0.69(33)	0.76(26)	0.72(30)
	1.95(0)	4.04	1.59(18)	1.80(8)	1.75(10)	0.57(0)	4.04	0.42(26)	0.45(20)	0.44(22)
	0.44(0)	11.57	0.36(18)	0.41(7)	0.39(11)	0.23(0)	11.57	0.19(21)	0.21(12)	0.20(15)
30m	1.96(0)	2.57	1.54(21)	1.74(11)	1.67(15)	0.78(0)	2.57	0.52(34)	0.55(30)	0.55(30)
	1.03(0)	4.04	0.80(22)	0.91(11)	0.86(14)	0.53(0)	4.04	0.38(28)	0.41(23)	0.40(24)
	0.34(0)	11.57	0.29(16)	0.32(6)	0.31(9)	0.21(0)	11.57	0.17(21)	0.19(19)	0.18(16)

One can observe that the difference between the FD soil models with variable water content is more pronounced as the rod length decreases and the water content increases. Considering the peak value of the transient GPR induced by the first return stroke for soil with $W = 2.57\%$ and electrode length of $L = 3$ m, it is reduced from 12.68 MV to 1.74 MV (reduced more than 7 times) when the rod length is increased by 10 times ($L = 30$ m). For the subsequent return stroke, it varies from 2.80 MV to 0.55 MV (reduced by more than 5 times) for the same variation in the electrode length. As seen in this table, the Smith-Longmire (SL) model has resulted in the lowest peak values for the transient GPR waveforms in all simulated cases. As detailed in Fig. 2a, the low-frequency resistivity provided by the SL model is lower compared with those obtained using Messier's (M) model and Scott's (S) model. As a result, GPR waveforms calculated using the SL model will present lower peak values than those assessed using the other models.

4.3. Time lag between the injected current and GPR

To investigate the time lag between the transient GPR and the injected current, the waveforms are plotted on the same graph as shown in Fig. 12. In this set of simulations, we have considered rod lengths of 3 m (short rod) and 30 m (long rod), frequency-independent (FI) soil (as it has been shown to be the most conservative scenario), soil resistivities of 100 and 1,000 Ωm, and a relative permittivity of 10. The GPR waveform is calculated for the first and subsequent lightning return strokes current, as presented by (12). In general, the GPR response waveforms

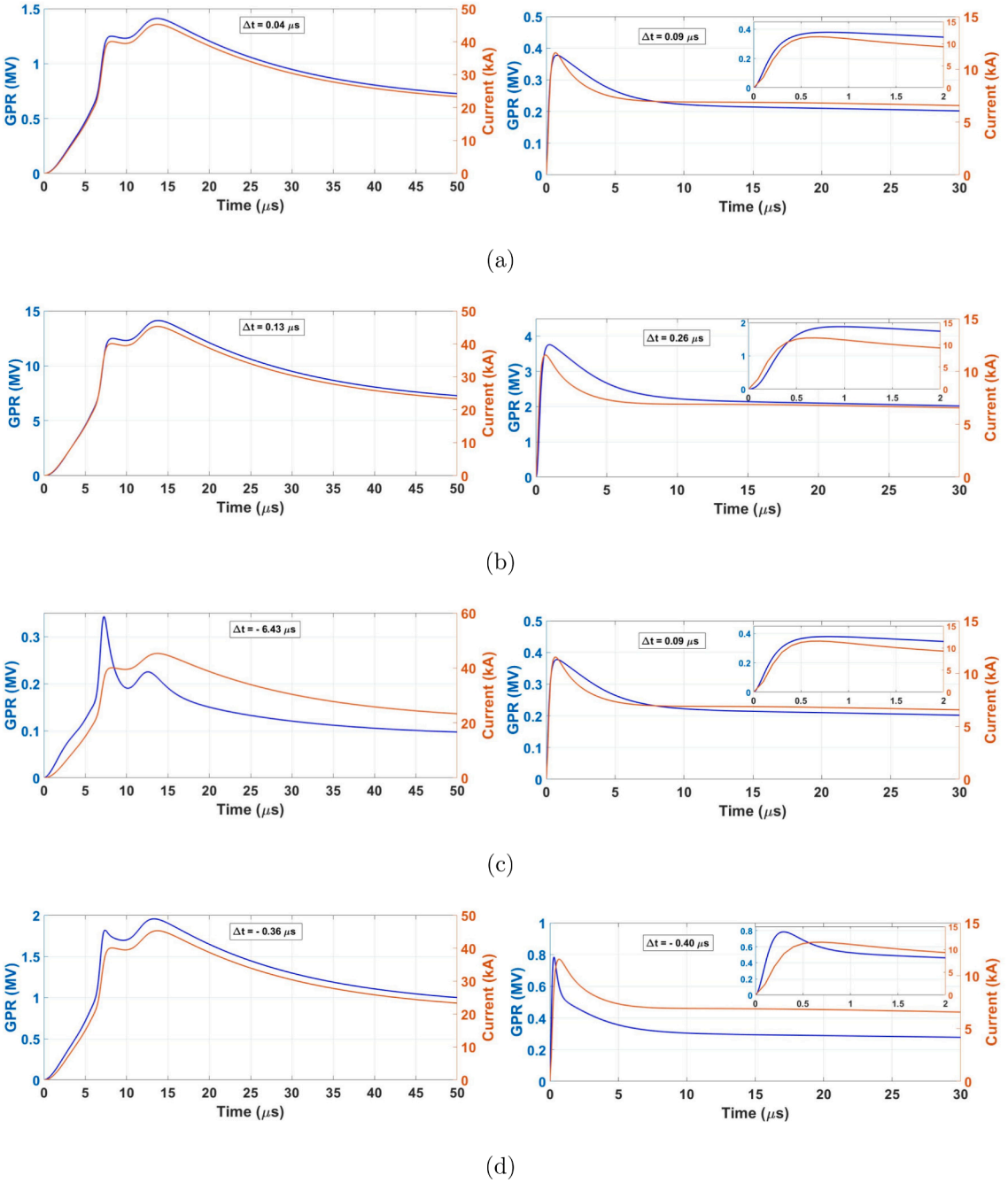


Fig. 12. Transient voltage responses (GPR) with injected current waveforms for the first return stroke (left-hand side column) and subsequent return stroke (right-hand side column) for: (a) $L = 3$ m and $\rho = 100$ Ωm, (b) $L = 3$ m and $\rho = 1{,}000$Ωm; (c) $L = 30$ m and $\rho = 100$Ωm and (d) $L = 30$ m and $\rho = 1{,}000$Ωm.

depends notably on the type of lightning current and the soil resistivity. According to Fig. 12a and b, concerning only the shorter electrode of 3 m, one notes that the time lag between the GPR and injected current waveforms is negligible for the first return stroke. In this case, the resistive behavior predominates, as shown in Fig. 5a and c (blue lines) up to 1 MHz. However, for the subsequent return stroke, which has a higher frequency content (due to its shorter front time, see Table 2) than the first stroke, the time lag becomes more relevant with increasing soil resistivity [44]. This occurs due to the predominant reactive part at higher frequencies, as seen in the harmonic impedance of the 3-m rod in Fig. 5a and c. As noted in Fig. 12a, the current is leading or lagging in relation to the voltage, due to the reactive part of the harmonic impedance of the electrode. Further, when the rod length is 30m, the developed GPR and the injected current have a pronounced time lag in the responses, see Fig. 12c and d, for both lightning currents. This occurs due to the predominance of the reactive impedance of the electrode's harmonic impedance, especially for frequencies above 100-kHz, as shown in the harmonic impedance of the 30-m rod in Fig. 7a and c. For high-resistive soils, Additionally, the lag time is more pronounced for the GPR developed by the subsequent return strokes associated with its higher frequency content due to the shorter front time of this lightning current, as depicted in Table 2. This can be seen in the $\Delta t = t_v - t_c$, where t_c and t_v are the times that occur the peaks of the current and voltage, respectively. The values of Δt are added in each sub-figure in Fig. 12.

4.4. Surge voltage across a string of insulators

The performance of the soil models in calculating the surge voltage

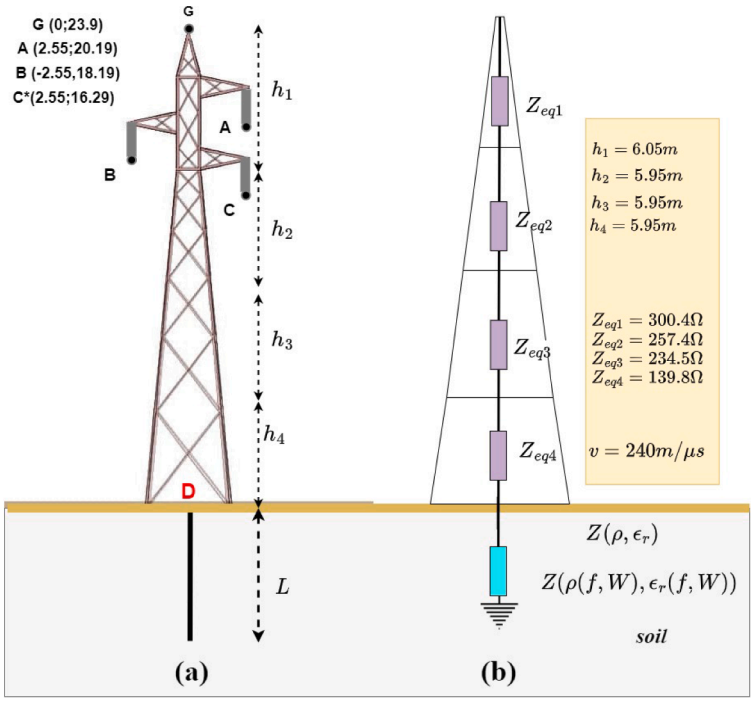


Fig. 13. 138-kV transmission tower: (a) Tower configuration (based on Fig. 3(a) in [1]); (b) Multi-conductor tower model.

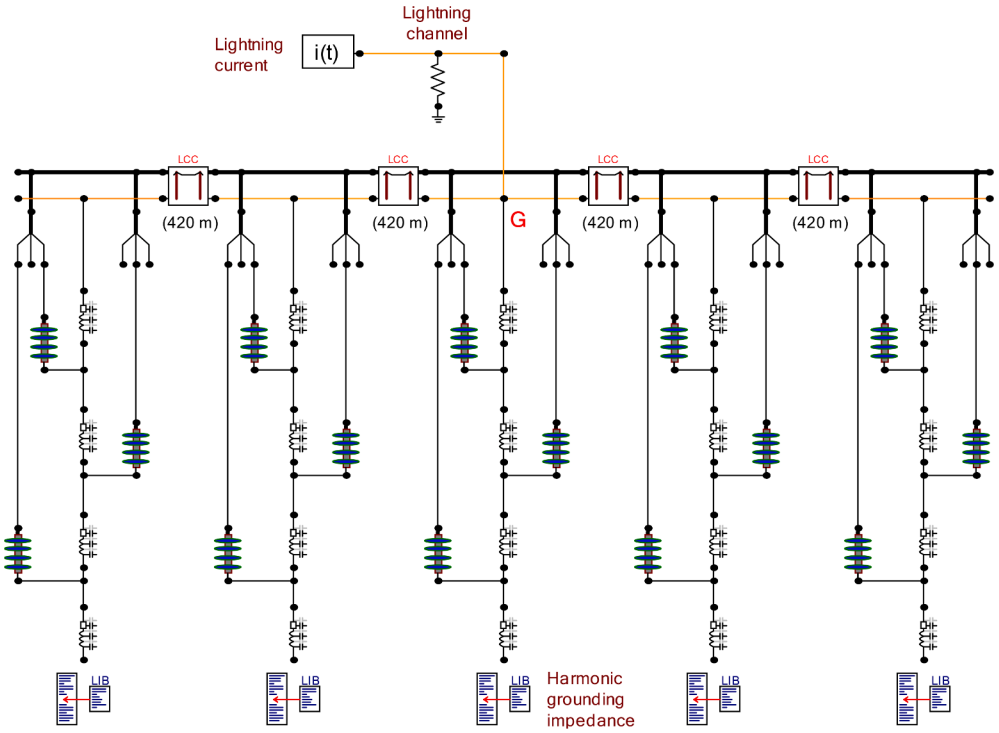


Fig. 14. Implemented circuit in ATP software to study the lightning performance.

(overvoltages) on a 138-kV transmission line (TL) tower subjected to the first and subsequent lightning strokes is investigated. For this purpose, the ATP software is utilized to carry out the time-domain simulations where the TL is located above a finitely-conducting soil. Similar to the previous section, three FD soil models with variable water content-Smith-Longmire (SL), Messier (M), and Scott (S) and frequency-independent soil were considered. Five transmission towers with four spans of 420 m are considered. The lightning strikes the central tower at its tip where the ground wire is connected, identified by the letter **G** in Fig. 13(a).

Using the ATP software, a multi-conductor tower model is used to simulate the tower, as shown in Fig. 13(b). The tower is modeled as four lossless short lines of length $(h_i), i = 1 - 4$ associated with a surge impedance Z_{eqi} calculated by Jordan's formula [45] with a propagation velocity v equal to 80% of the speed of light [46]. The line conductors are modeled by J. Marti's line model using the component LCC in the

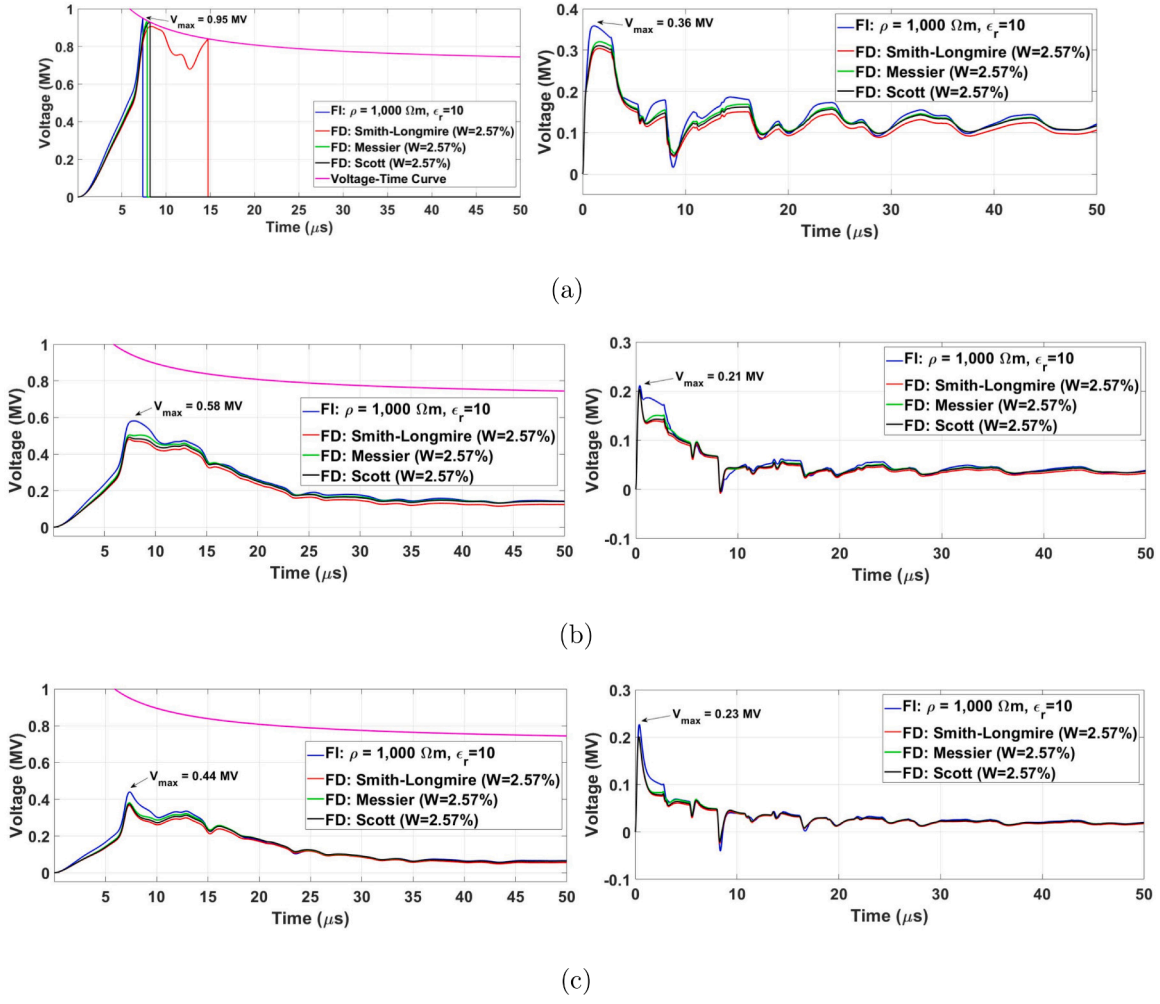


Fig. 15. Voltages across the upper string of insulators of the 138-kV tower induced by the first return stroke (shown on the left) and the subsequent return stroke (shown on the right) considering rod lengths of (a) $L = 3$ m, (b) $L = 15$ m, (c) $L = 30$ m.

ATP software. The TL is formed by phase conductors (A, B, and C) with a radius of 1.60 cm and per-unit-length (PUL) resistance of 0.063 Ω/km). The shield wire (G) has a radius of 0.80 cm and (PUL) resistance of 0.50 Ω/km. The equivalent circuit representation of the calculated HGI by means of rational approximations is obtained using the Vector Fitting technique, where poles and residues are determined. Then, the lumped elements associated with poles and residues are incorporated using the *lib component. The voltage across the upper phase string of insulators of the central tower is computed using the circuit configuration presented in [46]. Details of the string of insulators are found in [47]. The circuit representation is shown in Fig. 14.

If the voltage across the string of insulators is equal or exceeds the volt-time characteristics of the insulator, a backflashover occurs. The volt-time characteristic curve of an insulator is given by Zalhaf et al. [47]

$$V_F(t) = \left(K_1 + \frac{K_2}{t^{0.75}}\right)\ell \tag{14}$$

where K_1 and K_2 are coefficients, ℓ [m] is the length of the insulator chain, and t [μs] is the time after the lightning strikes at the tower. If the voltage across the string of insulators ever becomes equal to or greater than V_F at any instant, a backflashover (BF) occurs, resulting in the voltage across the string of insulators going to zero. The lightning strike is modeled by an impulse current source applied to the tower's tip, assuming a surge channel impedance of 400 − Ω. The insulator length ℓ is equal to 1.56 m (typical of 138 kV), $K_1 = 400$, and $K_2 = 710$ [47]. The

Table 4
Peak of voltages (in MV) across the upper string of insulators calculated for the frequency-independent (FI) and three frequency-dependent (FD) models of Smith-Longmire (SL), Messier (M), and Scott (S), with W=2.57% for different lengths of the rod L. The number in () is the percentage variation calculated using (13).

	Lightning First Return Stroke				Lightning Subsequent Return Stroke			
L (m)	FI soil	SL	M	S	FI soil	SL	M	S
3	0.95 (0)	0.91 (4)	0.94 (1)	0.93 (2)	0.36 (0)	0.32 (11)	0.34 (6)	0.33 (8)
15	0.57 (0)	0.47 (18)	0.50 (12)	0.49 (14)	0.21 (0)	0.20 (5)	0.20 (5)	0.20 (5)
30	0.43 (0)	0.36 (16)	0.38 (12)	0.37 (14)	0.23 (0)	0.20 (13)	0.20 (13)	0.20 (13)

voltage across the upper string of insulators induced by the return stroke current injected at the tip of the 138-kV tower with grounding rods of $L = 3$, and 30 m are arranged in Fig. 15. As seen, the voltage across the upper string of insulators with the FI soil model (blue line) in all cases presents a higher transient response than those computed assuming the FD soil models with variable water content as shown in Fig. 15,

Furthermore, as noted in Fig. 15 these induced voltage developed by the first stroke on the tower located above both types of soil exceeds the so-called volt-time characteristics curve (pink line) causing a back-flashover in this phase. However, the generated voltage across this string

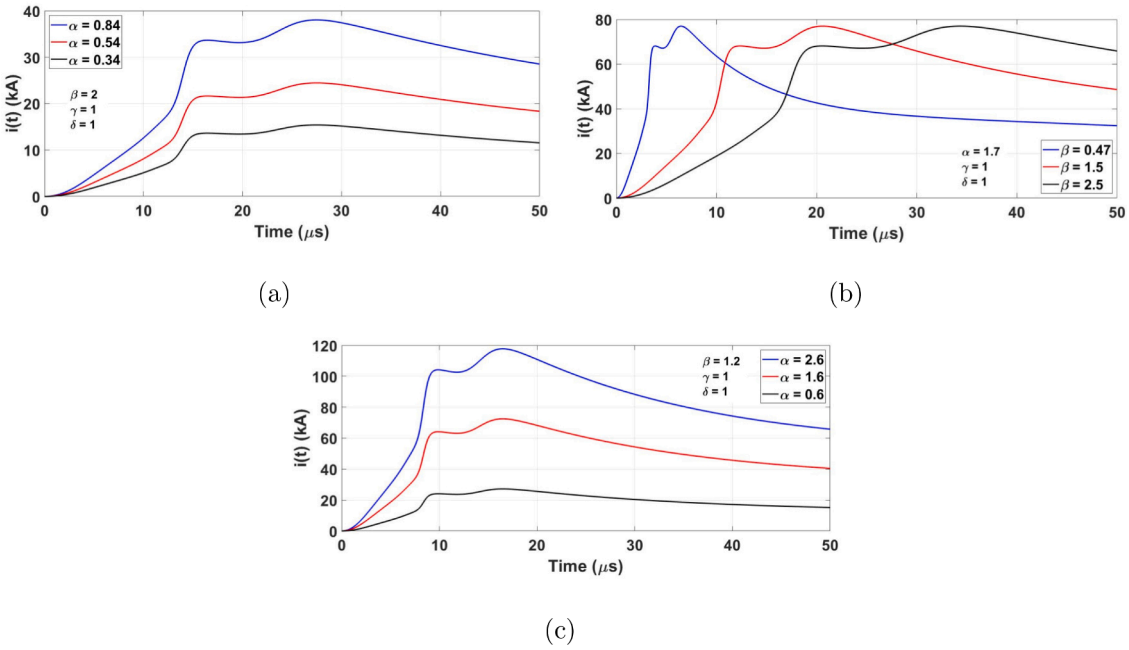


Fig. 16. Lightning current of first return stroke for scenarios: (a) S_1, (b) S_2 and (c) S_3.

for the current of a typical subsequent stroke has produced no backflashover for all cases. The steps to compute the overvoltages and the backflashover analysis are described in Fig. 4c.

One can see that as the length of the grounding rod increases, the peak of the voltage across the string of insulators decreases due to the reduction in the tower footing impedance. Using longer rods is an alternative to decrease these induced voltages across the string of insulators developed during the transient state [4]. Moreover, new topologies of grounding systems and the installation of line surge arresters may reduce the overvoltages and decrease the probability of the backflashover during the lightning strike [5].

To evaluate the performance of the three soil models, Smith-Longmire (SL), Messier (M), and Scott (S), the percentage variation $\delta(\%)$ between the peak of the transient voltage computed on the upper string of insulators are tabulated in Table 4. This table shows that the percentage variation of the voltage across the upper phase insulator is very low for the first return stroke. Furthermore, it is worth noting that the backflashover does not occur at the same time, especially for the Smith-Longmire (SL) model, which occurs at the second peak of the voltage waveform in Fig. 15a. Concerning the voltage across the upper string induced by the subsequent return stroke in Fig. 15b, one can see that the developed responses do not cause the string of insulators to flashover. On the other hand, a pronounced difference is observed in the voltage peaks when the frequency-independent soil model is employed in comparison with those under the assumption of a frequency-dependent soil model. The difference between the curves decreases as the rod length gets longer. These results confirm that the frequency effect and water content of soil electrical parameters must be incorporated in the computation of the lightning-induced surge in power systems. They influence the transient response, especially in the case of transmission towers located above highly-resistive soil (low water content) when subjected to fast-front lightning strikes. According to CIGRÉ [15], for studies involving TLs and grounding electrodes, the frequency effect on the soil electrical parameters must be included for soils above 700 Ωm to adequately calculate the transient responses involving practical engineering applications, as demonstrated in this work.

5. Backflashover analysis for distinct lightning waveforms

Finally, the impact of the three different FD soil models with variable water content on the backflashover analysis for distinct types of lightning currents is investigated in this section. For this purpose, the lightning current can be generically represented by a sum of m Heidler's functions given by Oliveira et al. [48]

$$i(t) = \sum_{k=1}^{m} \frac{I_{0k}}{\eta_k} \frac{(t/\tau_{1k})^{n_k}}{1 + (t/\tau_{1k})^{n_k}} e^{-t/\tau_{2k}} \tag{15a}$$

$$\eta_k = e^{\left[-\left(\frac{\tau_{1k}}{\tau_{2k}}\right)\left(n_k \frac{\tau_{2k}}{\tau_{1k}}\right)\right]^{1/n_k}} \tag{15b}$$

here I_{0k} is the current peak, τ_{1k} is the front-time constant, τ_{2k} is the decay-time constant, n_k is a coefficient related to the waveform steepness, and η_k is a peak correction factor. The lighting parameters for these both lightning currents are adopted from Moura et al. [48] assuming that $m = 7$, and the other parameters are organized as arrays with multiplying factors given by Oliveira et al. [48]

$$I_{0k} = \alpha[6 \quad 5 \quad 5 \quad 8 \quad 16.5 \quad 17 \quad 12 \times \delta][\text{kA}] \tag{16a}$$

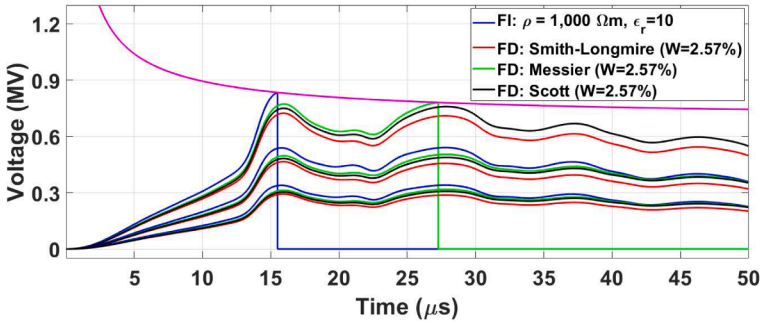
$$\tau_{1k} = \beta[3 \quad 3.5 \quad 4.8 \quad 6 \quad 7 \quad 70 \quad 12][\mu s] \tag{16b}$$

$$\tau_{2k} = \beta[76 \quad 10 \quad 30 \quad 26 \quad 23.2 \quad 200 \quad 26][\mu s] \tag{16c}$$

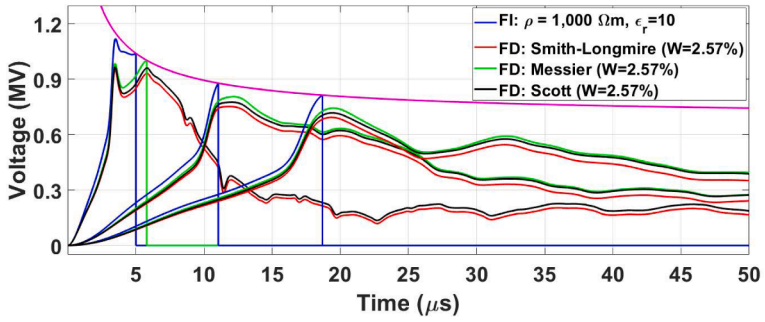
$$n_k = [2 \quad 3 \quad 5 \quad 9 \quad 30 \quad 2 \times \gamma \quad 14] \tag{16d}$$

where α, β, γ, and δ are the multiplying factor that adjusts the lightning waveforms (peak values I_{p1} and I_{p2}) and parameter T_{30} which is the time interval between 30% and 90% of the first peak (I_{p1}) on the front part, as detailed in [48]. In this simulations, three different scenarios **S** were considered:

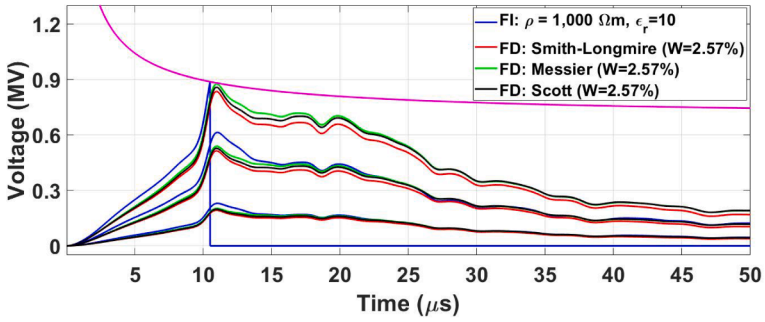
- S_1: Electrode length (L) of 3 m and lightning current $i(t)$ assuming three values of $\alpha = 0.84$, 0.54 and 0.34; $\beta = 2$, $\delta = \gamma = 1$;
- S_2: Electrode length (L) of 15 m and lightning current $i(t)$ assuming three values of $\alpha = 1.70$; $\beta = 0.47$, 1.5 and 2.5, $\delta = \gamma = 1$;
- S_3: Electrode length (L) of 30 m and lightning current $i(t)$ assuming three values of $\alpha = 2.6$, 1.6 and 0.6; $\beta = 1.20$, $\delta = \gamma = 1$;



(a)



(b)



(c)

Fig. 17. Voltages across the upper string of insulators of the 138-kV tower induced by the three different lightning currents for the scenarios: (a) S_1, (b) S_2 and (c) S_3.

The lightning current waveforms used in three different scenarios are plotted in Fig. 16. The same line configuration from Fig. 14 is adopted for each scenario. The generated voltages for the three distinct scenarios, considering the frequency-independent (FC) model, Smith-Longmire (SL), Messier (M), and Scott (S) models are plotted in Fig. 17.

According to this figure, the backflashover (BF) occurs at different instants depending on the lightning current waveform, soil model, and length of grounding rod (L). It can be seen that when the frequency-independent(FI) soil model is considered, the generated overvoltages are overestimated in comparison with the other frequency-dependent (FD) soil models. Further, the FC soil model produces overvoltages at the cross-arm of phase A that lead to a backflashover (BF) when high-amplitude current peaks are adopted in the three scenarios. In the scenarios S_1 and S_3, one notes that the BF also occurs when the Messier (M) model is used to represent the soil. In these scenarios, these BF simulations illustrate that peak values of the overvoltages across the string of insulators occur at different instants when compared with those generated for the FC soil models. These results indicate the importance of the soil models on the estimation of the overvoltages and prediction of the backflashover across the string of insulators, especially for those developed for high-peak associated with short-front time lightning current waveforms striking on power systems.

6. Conclusions

This paper studied the frequency effect and water content on the soil electrical parameters for different values of water content. An electromagnetic software FEKO was used to assess the harmonic impedance of grounding rods of $L = 3$ m, $L = 15$ m, and $L = 30$ m of length. The transient ground potential rise (GPR) of these rods and the induced voltage across the upper string of insulators were calculated. Results demonstrated that considering a frequency effect and variable water content, the harmonic impedance of the rods buried has significant reactive behavior (being either inductive or capacitive) depending on the frequency range. At low frequencies, the harmonic impedance is purely resistive, identified as static resistance. Considering the FD soil parameters, the harmonic impedance presents evident modifications, especially at high frequencies, compared with the FI soil model values.

Additionally, the low-frequency resistance R_{LF} presents a noticeable variation when specific FD soil models are used. The most considerable difference is obtained for the Smith-Longmire (SL) model due to the smallest low-frequency resistivity calculated by this approach. Furthermore, as the water content and the rod length increase, the difference in R_{LF} between the FD soil models reduces. The frequency effect on soil parameters provides a notable reduction in the transient GPR waveforms, especially at the peak values, compared with those obtained assuming the soil modeled by its frequency-independent parameters. These reductions are more significant for soils with low water content (highly-resistive ground). Additionally, the transient GPR waveforms developed by the subsequent return stroke have shown the highest percentage variation compared to those assessed induced by the first return stroke. Regarding the differences between the soil models, the Smith-Longmire model has presented the highest percentage variation in the transient GPR peaks for all rod lengths. This difference is caused by the higher frequency content of the subsequent return stroke due to its shorter front time.

Finally, the backflashover analysis has shown different voltage waveforms across the upper string of insulators using other soil models and lightning current characteristics (peak and front-time). It is demonstrated that the frequency-independent (FI) soil model overestimates the voltages across the string of insulators due to its higher soil resistivity. The Smith-Longmire (SL) model has presented the lowest voltage waveform, which impacts the occurrence of the backflashover at a later time than those flashovers calculated using the Scott (S) or Messier (M) model. Longer rods have lower grounding impedance and provide a reduction in the voltage peak across the string of insulators developed by lightning strikes. However, depending on the lightning waveform and soil model, the backflashover may occur at different instants during the transient state. These results indicated the importance of the proper modeling of the soil to estimate the overvoltages across the string of insulators and predict the backflashover occurrence.

Additionally, distinct topologies of tower-footing grounding systems for achieving low harmonic impedance and installing line surge arresters are alternatives to reduce the overvoltages across the string of insulators, decreasing the probability of backflashover and outages during the transient state. As contributions, the impact of different soil modeling with water content on overvoltages generated by lightning strikes on transmission lines was demonstrated. These overvoltages can cause backflashover depending on the soil model adopted.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Data availability

Data will be made available on request.

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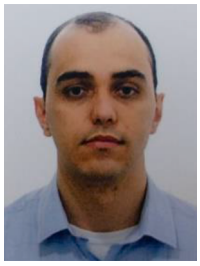
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